

Fundamentals (15 Questions)

1. What is Generative AI and how does it differ from traditional AI?
2. Explain the concept of Large Language Models (LLMs).
3. What is the transformer architecture and why is it important?
4. Define prompt engineering and its significance.
5. What is the difference between supervised and unsupervised learning in GenAI?
6. Explain the concept of tokenization in language models.
7. What are embeddings and how are they used in GenAI?
8. Define fine-tuning in the context of language models.
9. What is the difference between GPT and BERT models?
10. Explain the concept of attention mechanism in transformers.
11. What is the purpose of positional encoding in transformers?
12. Define autoregressive language modeling.
13. What is the difference between encoder-only and decoder-only architectures?
14. Explain the concept of few-shot learning.
15. What is the role of pre-training in language models?

Basic Implementation (15 Questions)

16. How would you use OpenAI's API to generate text?
17. What is LangChain and what are its main components?
18. How do you implement a simple chatbot using GPT?
19. What is the difference between temperature and top_p parameters?
20. How would you handle API rate limits in GenAI applications?
21. What is prompt chaining and how do you implement it?
22. How do you implement conversation memory in a chatbot?
23. What is the purpose of system prompts in chat models?
24. How would you implement text summarization using GenAI?
25. What is the difference between completion and chat completion APIs?
26. How do you implement content filtering in GenAI applications?
27. What is the purpose of max_tokens parameter?
28. How would you implement a simple RAG system?
29. What is the difference between streaming and non-streaming responses?
30. How do you handle errors in GenAI API calls?

Basic Concepts (10 Questions)

31. What is hallucination in GenAI and how can it be reduced?
32. Explain the concept of bias in language models.
33. What is the difference between zero-shot and few-shot prompting?
34. Define chain-of-thought prompting.
35. What is the purpose of role-based prompting?

36. Explain the concept of prompt injection attacks.
37. What is the difference between instruction tuning and fine-tuning?
38. Define few-shot learning with examples.
39. What is the purpose of human feedback in RLHF?
40. Explain the concept of model alignment.

Practical Applications (10 Questions)

41. How would you build a customer service chatbot?
 42. What are the key considerations for building a content generation tool?
 43. How would you implement a code generation assistant?
 44. What is the best approach for building a document Q&A system?
 45. How would you create a text summarization tool?
 46. What are the steps to build a language translation system?
 47. How would you implement a creative writing assistant?
 48. What is the approach for building a code review tool?
 49. How would you create a personal assistant using GenAI?
 50. What are the considerations for building a learning platform with GenAI?
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Mid-Level - 60 Questions

Advanced Architecture (15 Questions)

51. Explain the differences between GPT, LLaMA, and Claude architectures.
52. What is Mixture of Experts (MoE) and how does it work?
53. Explain the concept of sparse attention patterns.
54. What is the difference between LoRA and full fine-tuning?
55. How does QLoRA work and what are its benefits?
56. Explain the concept of model parallelism in large language models.
57. What is the purpose of gradient checkpointing?
58. How does flash attention improve efficiency?
59. Explain the concept of model quantization.
60. What is the difference between dynamic and static quantization?
61. How does knowledge distillation work in language models?
62. Explain the concept of model pruning.
63. What is the purpose of model compression techniques?
64. How does speculative decoding work?
65. Explain the concept of model serving optimization.

RAG and Retrieval (15 Questions)

66. Design a RAG system for 1 million documents.
67. What are the different chunking strategies for RAG?
68. How do you implement hybrid search in RAG systems?
69. What is the difference between dense and sparse retrieval?
70. How do you implement reranking in RAG systems?
71. What are the best practices for vector database selection?
72. How do you handle metadata filtering in RAG?
73. What is the purpose of query expansion in RAG?
74. How do you implement multi-hop reasoning in RAG?
75. What are the challenges of RAG in production?
76. How do you evaluate RAG system performance?
77. What is the difference between RAG and fine-tuning?
78. How do you implement iterative retrieval?
79. What are the best practices for RAG prompt engineering?
80. How do you handle RAG system scalability?

System Design (15 Questions)

81. Design a scalable GenAI system for 100,000 concurrent users.
82. How do you implement caching in GenAI applications?
83. What is the best approach for model versioning?
84. How do you implement A/B testing for GenAI models?
85. What are the considerations for multi-tenant GenAI systems?
86. How do you implement load balancing for GenAI services?
87. What is the approach for handling model drift?
88. How do you implement circuit breakers in GenAI systems?
89. What are the best practices for error handling in GenAI?
90. How do you implement monitoring for GenAI applications?
91. What is the approach for cost optimization in GenAI?
92. How do you implement security in GenAI systems?
93. What are the considerations for data privacy in GenAI?
94. How do you implement backup and recovery for GenAI?
95. What is the approach for disaster recovery in GenAI?

Advanced Implementation (15 Questions)

96. How do you implement multi-agent systems with GenAI?
97. What is the approach for building autonomous agents?
98. How do you implement tool calling in language models?
99. What are the best practices for function calling?
100. How do you implement memory management in agents?
101. What is the approach for building reasoning agents?
102. How do you implement planning in AI agents?
103. What are the considerations for agent communication?

- 104. How do you implement agent coordination?
 - 105. What is the approach for building multi-modal agents?
 - 106. How do you implement agent learning and adaptation?
 - 107. What are the best practices for agent evaluation?
 - 108. How do you implement agent safety mechanisms?
 - 109. What is the approach for building explainable agents?
 - 110. How do you implement agent debugging and monitoring?
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Senior Level - 60 Questions

Architecture and Design (20 Questions)

- 111. Design a multi-modal GenAI system that processes text, images, and audio.
- 112. How do you architect a GenAI system for enterprise deployment?
- 113. What is the approach for building GenAI systems with microservices?
- 114. How do you design GenAI systems for edge deployment?
- 115. What are the considerations for building GenAI systems in the cloud?
- 116. How do you architect GenAI systems for real-time applications?
- 117. What is the approach for building GenAI systems with high availability?
- 118. How do you design GenAI systems for global deployment?
- 119. What are the considerations for building GenAI systems with compliance?
- 120. How do you architect GenAI systems for multi-region deployment?
- 121. What is the approach for building GenAI systems with data governance?
- 122. How do you design GenAI systems for audit and compliance?
- 123. What are the considerations for building GenAI systems with security?
- 124. How do you architect GenAI systems for privacy protection?
- 125. What is the approach for building GenAI systems with ethical AI?
- 126. How do you design GenAI systems for bias mitigation?
- 127. What are the considerations for building GenAI systems with fairness?
- 128. How do you architect GenAI systems for transparency?
- 129. What is the approach for building GenAI systems with accountability?
- 130. How do you design GenAI systems for human oversight?

Leadership and Strategy (20 Questions)

- 131. How do you lead a GenAI transformation in a Fortune 500 company?
- 132. What is your approach to building a GenAI center of excellence?
- 133. How do you develop GenAI strategy for different business units?
- 134. What is the approach for securing executive buy-in for GenAI?
- 135. How do you manage GenAI projects with multiple stakeholders?
- 136. What are the considerations for GenAI talent acquisition?

- 137. How do you build and lead GenAI teams?
- 138. What is the approach for GenAI team development and training?
- 139. How do you manage GenAI vendor relationships?
- 140. What are the considerations for GenAI partnership strategies?
- 141. How do you handle GenAI project failures and recovery?
- 142. What is the approach for GenAI risk management?
- 143. How do you manage GenAI compliance and regulatory requirements?
- 144. What are the considerations for GenAI intellectual property?
- 145. How do you handle GenAI competitive analysis?
- 146. What is the approach for GenAI market research?
- 147. How do you manage GenAI customer relationships?
- 148. What are the considerations for GenAI product management?
- 149. How do you handle GenAI sales and marketing?
- 150. What is the approach for GenAI business development?

Advanced Technical (20 Questions)

- 151. How do you implement advanced prompt engineering techniques?
- 152. What is the approach for building custom language models?
- 153. How do you implement advanced fine-tuning strategies?
- 154. What are the considerations for building domain-specific models?
- 155. How do you implement advanced evaluation methodologies?
- 156. What is the approach for building automated evaluation systems?
- 157. How do you implement advanced monitoring and observability?
- 158. What are the considerations for building GenAI MLOps?
- 159. How do you implement advanced security measures?
- 160. What is the approach for building GenAI threat detection?
- 161. How do you implement advanced privacy protection?
- 162. What are the considerations for building GenAI data protection?
- 163. How do you implement advanced bias detection and mitigation?
- 164. What is the approach for building GenAI fairness systems?
- 165. How do you implement advanced explainability techniques?
- 166. What are the considerations for building GenAI interpretability?
- 167. How do you implement advanced human-AI collaboration?
- 168. What is the approach for building GenAI human-in-the-loop systems?
- 169. How do you implement advanced agent architectures?
- 170. What are the considerations for building GenAI autonomous systems?

Expert Level - 30 Questions

Research and Innovation (15 Questions)

- 171. What are the latest advances in transformer architectures?
- 172. How do you contribute to GenAI research and development?
- 173. What is the approach for building next-generation language models?
- 174. How do you implement cutting-edge GenAI techniques?
- 175. What are the considerations for GenAI research ethics?
- 176. How do you handle GenAI research collaboration?
- 177. What is the approach for GenAI patent development?
- 178. How do you implement GenAI research validation?
- 179. What are the considerations for GenAI research publication?
- 180. How do you handle GenAI research funding?
- 181. What is the approach for GenAI research team building?
- 182. How do you implement GenAI research infrastructure?
- 183. What are the considerations for GenAI research data?
- 184. How do you handle GenAI research partnerships?
- 185. What is the approach for GenAI research commercialization?

Industry Leadership (15 Questions)

- 186. How do you shape GenAI industry standards and best practices?
 - 187. What is the approach for GenAI thought leadership?
 - 188. How do you influence GenAI policy and regulation?
 - 189. What are the considerations for GenAI industry collaboration?
 - 190. How do you handle GenAI industry conferences and events?
 - 191. What is the approach for GenAI industry networking?
 - 192. How do you implement GenAI industry partnerships?
 - 193. What are the considerations for GenAI industry competition?
 - 194. How do you handle GenAI industry disruption?
 - 195. What is the approach for GenAI industry transformation?
 - 196. How do you implement GenAI industry innovation?
 - 197. What are the considerations for GenAI industry sustainability?
 - 198. How do you handle GenAI industry challenges?
 - 199. What is the approach for GenAI industry opportunities?
 - 200. How do you implement GenAI industry vision and strategy?
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